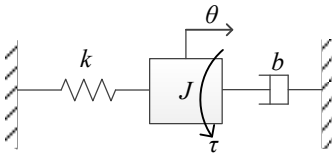
Huazhong University of Science and Technology
School of Mechanical Science and Engineering
System Dynamics

1. Consider a mass-spring-damper system as shown in Fig. 1. Derive the equation of motion. For the following inputs and initial conditions, take Laplace Transform and find Θ(s); derive the rotational displacement response θ(t) with Inverse Laplace Transform::



a) τ = unit step, $\theta(0)=\dot{\theta}(0)=0$, J = 1, b = 3, k = 2.

b) $\tau=0, \theta(0)=1, \dot{\theta}(0)=0$, J = 1, b = 3, k = 2.

Hint: For $\Theta(s)=L[\theta(t)]$, the Laplace Transform of its derivatives can be obtained as

$$L\left[\ddot{\theta}(t)\right]=s^2\Theta(s)-s\theta(0)-\theta'(0),\quad L\left[\dot{\theta}(t)\right]=s\Theta(s)-\theta(0)$$

Solution: Equation of motion $J\ddot{\theta}+b\dot{\theta}+k\theta=\tau$

The transfer function $\dfrac{\Theta(s)}{\Gamma(s)}=\dfrac{1}{Js^2+bs+k}$

$$L\left[\ddot{\theta}(t)\right]=s^2\Theta(s)-s\theta(0)-\theta'(0)$$
$$L\left[\dot{\theta}(t)\right]=s\Theta(s)-\theta(0)$$

(a) $\Gamma(s)=\dfrac{1}{s}, \theta(0)=\dot{\theta}(0)=0$, taking Laplace Transform of the equation of motion

$$\left[s^2\Theta(s)-s\theta(0)-\theta'(0)\right]+3\left[s\Theta(s)-\theta(0)\right]+2\Theta(s)=\frac{1}{s}$$

$$\Rightarrow\Theta(s)=\frac{1}{s\left(s^2+3s+2\right)}=\frac{0.5}{s}-\frac{1}{s+1}+\frac{0.5}{s+2}\Rightarrow\theta(t)=0.5-e^{-t}+\frac{1}{2}e^{-2t}$$

(b) $\Gamma(s)=0, \theta(0)=1, \dot{\theta}(0)=0$, taking Laplace Transform of the equation of motion

$$\left[s^2\Theta(s)-s\theta(0)-\theta'(0)\right]+3\left[s\Theta(s)-\theta(0)\right]+2\Theta(s)=0$$

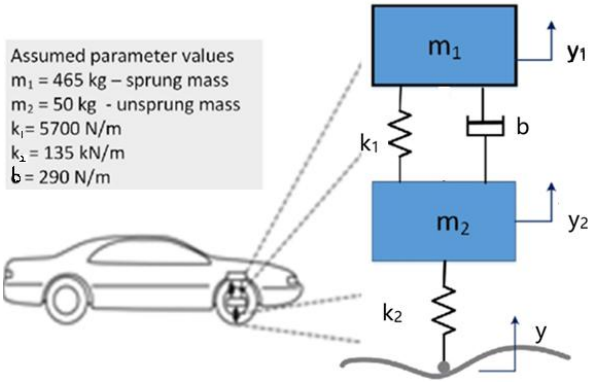
$$\Rightarrow\Theta(s)=\frac{s+3}{s^2+3s+2}=\frac{2}{s+1}-\frac{1}{s+2}\Rightarrow\theta(t)=2e^{-t}-e^{-2t}$$

2. Linearize $y=\sin(\theta)+\cos(\theta)+\theta\cos(\theta)+\dot{\theta}\sin(\theta)$ around the equilibrium $\theta=\dot{\theta}=0$

Solution: $y=f(0,0)+\left[\cos(\theta)-\sin(\theta)+\cos(\theta)-\theta\sin(\theta)+\dot{\theta}\cos(\theta)\right]\Big|_{0,0}\cdot(\theta-0)+\left[\sin(\theta)\right]\Big|_{0,0}\cdot(\dot{\theta}-0)=1+2\theta$

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3. Derive the dynamic equation of the 1/4 Car Model and find the response of $y_1(t)$ for a step change in y.



Solution:

Step 1: Equations of Motion

Using Newton's second law around equilibrium (gravity eliminated):

For m_1:

$$m_1\ddot{y}_1 = k_1(y_2 - y_1) + b(\dot{y}_2 - \dot{y}_1) \Rightarrow m_1\ddot{y}_1 + b\dot{y}_1 + k_1y_1 = b\dot{y}_2 + k_1y_2$$

For m_2:

$$m_2\ddot{y}_2 = k_2(y-y_2) - k_1(y_2-y_1) - b(\dot{y}_2-\dot{y}_1) \Rightarrow m_2\ddot{y}_2 + b\dot{y}_2 + (k_1+k_2)y_2 = b\dot{y}_1 + k_1y_1 + k_2y$$

Step 2: Laplace Transform (Zero Initial Conditions)

Let $Y_1(s), Y_2(s), Y(s)$ be Laplace transforms.

From m_1 equation:

$$(m_1s^2 + bs + k_1)Y_1 = (bs + k_1)Y_2 \quad (1)$$

From m_2 equation:

$$[m_2s^2 + bs + (k_1 + k_2)]Y_2 = (bs + k_1)Y_1 + k_2Y \quad (2)$$

Step 3: Transfer Function $G(s) = \frac{Y_1(s)}{Y(s)}$

Solve (1) for Y_2:

$$Y_2 = \frac{m_1s^2 + bs + k_1}{bs + k_1}Y_1$$

Substitute into (2):

$$\left[m_2s^2 + bs + k_1 + k_2\right] \cdot \frac{m_1s^2 + bs + k_1}{bs + k_1}Y_1 = (bs + k_1)Y_1 + k_2Y$$

Multiply both sides by $(bs + k_1)$:

$$\left[m_2s^2 + bs + k_1 + k_2\right](m_1s^2 + bs + k_1)Y_1 = (bs + k_1)^2Y_1 + k_2(bs + k_1)Y$$

Rearrange:

$$\left\{[m_2s^2 + bs + k_1 + k_2](m_1s^2 + bs + k_1) - (bs + k_1)^2\right\}Y_1 = k_2(bs + k_1)Y$$

Thus:

$$G(s) = \frac{Y_1(s)}{Y(s)} = \frac{k_2(bs + k_1)}{[m_2s^2 + bs + k_1 + k_2][m_1s^2 + bs + k_1] - (bs + k_1)^2}$$

Step 4: Substitute Numerical Values

Numerator:

$$k_2(bs + k_1) = 135000 \cdot (290s + 5700) = 39150000s + 769500000$$

Denominator (after expansion and simplification):

$$D(s) = 23250s^4 + 149350s^3 + 65710500s^2 + 39150000s + 769500000$$

So:

$$G(s) = \frac{39150000s + 769500000}{23250s^4 + 149350s^3 + 65710500s^2 + 39150000s + 769500000}$$

With input $Y(s) = \frac{1}{s}$, we get:

$$Y_1(s) = G(s) \cdot \frac{1}{s} = \frac{39150000s + 769500000}{s \cdot D(s)}$$